Reproducibility Audit Report

Deliverable 06 · Research Methods and Scientific Integrity in AI · UNMSM 2026-II · Deisy Lizbeth Acosta Ticse Instrument: Session 6 reproducibility scorecard (9 items; Yes / Partial / No). Audit date: 25 Oct 2026.

1. Paper metadata

Hou, X., Liu, Z., Wang, G., & Shi, N. (2026). Bridging awareness and behavior: decoding implicit metacognitive behaviors in AI-assisted programming via fine-grained log analysis. *Frontiers in Psychology, 17*. https://doi.org/10.3389/fpsyg.2026.1790196 (open access).

Why this paper. It is one of the 10 studies included in my Deliverable 04 review (record R047), it reports a randomised trial of an AI scaffolding agent, and its headline result is a single post-test difference.

2. Experimental setup (as reported)

Element	Reported
Population	126 undergraduates recruited (convenience sampling), one Python course, one Chinese university; 122 analysed (attrition 3.2 %, by group not reported)
Allocation	Individual randomisation with Excel `=RAND()`; AI n = 62, control n = 60; allocation concealment not described
Intervention	AI agent in a customised Jupyter Notebook; rule-based triggers (e.g., first keystroke < 60 s, idle > 10 min) launch non-directive metacognitive prompts; example prompts in Table 1; **underlying LLM not named**
Control	Native Jupyter Notebook, no AI
Outcomes	Post-test programming score (isomorphic task; rubric; two blinded raters; ICC = 0.81 on a 30 % subset); modified MAI (self-report); 11 log-derived behaviour metrics
Analysis	t-tests / Mann–Whitney U; linear mixed model for MAI; Bonferroni for the 11 log metrics; Shapiro–Wilk, Q–Q plots, Levene

3. Scorecard

#	Item	Rating	Evidence and reasoning
1	Seeds reported, many	**No**	The `=RAND()`

#	Item	Rating	Evidence and reasoning
	runs?		allocation has no recorded seed, so the allocation cannot be regenerated. A single trial; no replication.
2	Splits described? (here: group allocation)	**Partial**	Method and group sizes given; concealment, attrition by group and age/gender balance not reported.
3	Leakage checked? (here: contamination and outcome overlap)	**Partial**	Raters blinded (good). No safeguards against information flowing between groups in the same course; the post-test is isomorphic to practice tasks, so it measures near transfer only.
4	Tuning separated? (here: trigger thresholds)	**Partial**	Thresholds are called “predefined”, but their origin is not stated; it is unclear whether they were set before or after seeing this cohort’s logs.
5	Statistical test named?	**Yes**	Tests, assumption checks and correction for the log metrics are named.
6	Effect size and CI?	**Partial**	Cohen’s d and r reported; **no confidence intervals**; the headline d is internally inconsistent (Section 4).

#	Item	Rating	Evidence and reasoning
7	External validation?	**No**	One site, one course, one cohort; authors acknowledge limited generalisability.
8	Compute documented?	**No**	LLM identity and version, Python/SciPy/Pandas versions, Jupyter extension and hardware are not reported.
9	Code and data available?	**Partial**	Data "available by the authors, without undue reservation" (on request); no code or full prompt set released.

Overall score: 7 / 18 (39 %) — Yes = 2, Partial = 1, No = 0. **Judgement: low reproducibility.** The design is sound in principle (randomised, blinded scoring, baseline equivalence tested), but a stranger could not rebuild the intervention (unknown model, partial prompts, no code) or regenerate the allocation, and the reported statistics contain inconsistencies.

4. Statistical rigor — numbers I recomputed from the reported values

Reported	Recomputed	Problem
Post-test 85.2 (SD 9.4, n = 62) vs 76.5 (SD 10.8, n = 60); t(120) = 4.75, **p = 0.002** (Section 5.2)	t = 4.75 gives **p ≈ 0.000006**; Table 7 itself says p < 0.001	Text and table disagree
d = 0.76	Pooled SD 10.11 → **d = 0.86**; the reported t is consistent with 0.86, not 0.76	Effect size misreported
No CI	Mean difference 8.7, **95 % CI 5.1 to 12.3** (from reported means/SDs)	CI should be reported
Pre-test MAI covariate β = 0.044, SE = 0.012, **p = 0.174** (Table 8)	β/SE = 3.67 → **p ≈ 0.0002**	Coefficient, SE and p cannot all be correct
Control MAI 51.9 + 10.9	= 62.8 (reported 62.5)	Arithmetic slip
Two "pre-test" programming measures (Table 6: 52.7 vs 51.0; Table 7: 68.4 vs 67.9)	—	Not explained

Other concerns: very large effects on log metrics (d = 4.08 and 3.87) from a 21-day intervention warrant scrutiny of metric definitions; no power analysis; multiple-comparison correction applied only to the log metrics, not to all outcome families; recruitment is dated January–February 2025 while log timestamps are from May 2025.

Exit-ticket number: the headline *p = 0.002 with d = 0.76* is not a lucky run but a **reporting error**: the reported data imply a larger, more certain effect (d = 0.86, p < 0.001). The error does not overturn the finding, but it shows the numbers were not checked against each other.

5. Missing information

LLM name and version; full prompt set and trigger code; randomisation seed and concealment; power analysis; attrition by group; confidence intervals; software and hardware versions; analysis scripts; anonymised log data in a repository; a delayed post-test without the AI (the authors themselves note persistence after removal is unknown).

6. Recommendations (three fixes that would help most)

1. **Release the intervention:** name the LLM and version, publish the full prompt set, trigger rules and Jupyter extension code, plus anonymised logs and analysis scripts with pinned versions.
2. **Correct and complete the statistics:** reconcile the p-value, recompute d, fix Table 8, report confidence intervals for every primary effect, and state the multiple-comparison strategy for all outcome families.
3. **Strengthen the design for replication:** pre-register with a power analysis, conceal allocation (seeded, documented), add a delayed post-test without the AI, and replicate in a second course or institution.

7. Relevance to my thesis

This study is evidence in my Deliverable 04 review. The audit lowers the weight I give it (already flagged there as “LLM not specified”) and turns its gaps into requirements for my own protocol: name the model and version, publish prompts, report CIs, and include an unassisted post-test.

Verification note: values were extracted from the HTML full text and checked by recomputation; tables should be re-checked against the PDF version of record before citing. AI-use disclosure: Claude (Anthropic) assisted in extracting reported values, recomputing statistics and drafting the report; ratings and judgement reviewed by the author.